Jay Doshi

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Education

B.Sc. in Computer Science & Engineering **GPA: 3.7/4.0**
University of Toledo, Toledo, OH Expected Graduation: December 2027

Relevant Coursework: Kalman Filtering, Probability and Statistics, Stochastic Processes, Linear Algebra, Multivariable Calculus, Artificial Intelligence

Experience

Network Engineer Intern | Cenovus Energy, Oregon, OH **September 2026 – Present**

- Collaborated with OT and process-control engineers to validate system integrations and support production cutovers.
- Analyzed SolarWinds NPM metrics to perform root-cause analysis, resolving performance issues before production outages.
- Integrated server, VMware virtualized, and backup systems into stable hosting environments supporting plant operations.

Process Controls Intern | Cleveland-Cliffs, Coatesville, PA **May 2026 – August 2026**

- Engineered threshold-based anomaly detection comparing live gauge data to plate specs, cutting defective plate escapes 30%.
- Built ETL pipelines moving live OpenVMS sensor data into SQL Server for process monitoring, cutting manual effort 40%.
- Increased engineering and business decision efficiency by 50% with Power BI dashboards on SQL views and stored procedures.
- Deployed VB.NET journaling microservices to a clustered QMS app, restoring TCP/IP data flow between PLCs and SQL Server.

IT Support Technician Level 2 | University of Toledo, Toledo, OH **September 2025 – May 2026**

- Resolved 40+ monthly IT tickets by troubleshooting Windows, hardware, software, and network issues, minimizing downtime.
- Imaged and deployed 150+ Windows devices using CLI tools for BIOS updates, data backups, and software installation.

Undergraduate Research Assistant | University of Toledo, Toledo, OH **May 2025 – December 2025**

- Reduced target navigation error by 40% by iteratively tuning PID gains against setpoint and response curves in QGroundControl.
- Prototyped PID control designs in MATLAB, trialing gain settings for the PX4 drone flight controller.
- Developed a ROS2-Gazebo digital twin of the Holybro X500 drone to simulate and validate PX4 control logic before flights.
- Improved control accuracy by 60% by quantifying RMSE and overshoot in Python (Pandas, NumPy) across 15+ flight trials.
- Designed and evaluated a PPO-based reinforcement learning model for autonomous drone target navigation in ROS2-Gazebo.

Professional Projects

SportsSight Analytics | Python, K-Means, YOLOv8, OpenCV, NumPy, Matplotlib **August 2025 – December 2025**

- Fine-tuned YOLOv8 on Roboflow football dataset, achieving 90% player and ball detection accuracy on test video clips.
- Applied K-Means pixel clustering for automated team assignment by t-shirt color, improving segmentation accuracy to 75%.
- Implemented optical flow and perspective transformation to compute real-world player speed and distance in meters.
- Generated real-time match analytics on ball possession and movement patterns to support data-driven sports insights.

DeepTruth | Python, PyTorch, DistilBERT, Gemini API, AWS, GCP, MongoDB, React **January 2025 – March 2025**

- Fine-tuned DistilBERT on 9.5K+ labeled records, enabling dual-model scoring with 70% Gemini and 30% BERT weighting.
- Built Python ETL pipeline with automated validation checks, improving data ingestion accuracy across 9,500+ records.
- Scheduled Transformer retraining with AWS Lambda and CloudWatch cron workflows for ML lifecycle management.
- Won RocketHacks 2025 for developing scalable AI fact-checking product used by 1,000+ users over a 6-month period.

SkillBridge Learning Platform | NLP, TF-IDF, PostgreSQL, Django, AWS **August 2024 – December 2024**

- Engineered NLP-based recommendation engine using TF-IDF and cosine similarity, achieving 95% prediction accuracy in testing.
- Designed PostgreSQL schemas with dimensional modeling, improving query performance and reporting speed by 80%.
- Developed Gemini API-powered AI agent auto-generating video transcripts and MCQs, reducing manual content effort.

Technical Skills

Programming Languages: Python, MATLAB, SQL, C++, C, Java, R, JavaScript

Controls & Simulation: PID Control, Digital Twin, ROS2, Gazebo, PX4, QGroundControl, Reinforcement Learning

Machine Learning: Supervised/Unsupervised Learning, Anomaly Detection, Computer Vision, NLP, Deep Learning, MLOps

Data & Visualization: Pandas, NumPy, Matplotlib, Power BI Dashboards, Tableau, Excel (VLOOKUP, PivotTables)

Databases & Tools: SQL Server, PostgreSQL, MySQL, MongoDB, Git, Linux Command Line, AWS, Google Cloud Platform